

# Muhammad Usman Ashraf (Ph.D)

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**Target Role:** Detector Developer Software, with secondary fit for Muon Detector Developer Hardware.

Detector R&D Scientist | Tracking Software | GEANT4 Simulation | Physics-Informed ML

## Profile Summary

*PhD experimental physicist with 6 years of postdoctoral experience in detector R&D, tracking detector operation, reconstruction software, GEANT4 simulation, data quality monitoring, calibration, and large-scale scientific data analysis. Experienced across sPHENIX, STAR, NA62, CMS, and ALICE, with hands-on work in TPC tracking calibration, GTK beam spectrometer commissioning, silicon sensor testing, detector performance validation, Monte Carlo production, and analysis software development. Strong fit for muon detector software development, multi-track reconstruction, physics-informed machine learning, detector response simulation, and validation workflows. Proven track record for proposal development to funding agencies for successful awards*

## Relevant Detector, Tracking, and Software Skills

- Machine Learning and Data Analysis: supervised learning, feature engineering, model validation, performance metrics, scientific ML, large-scale data analysis, statistical modeling.
- Tracking and Reconstruction: charged particle tracking, vertex reconstruction, track quality selection, detector alignment and calibration support, pileup aware event analysis, strange and neutral hadron reconstruction, particle correlation workflows.
- Detector R&D and Commissioning: TPC calibration, GTK beam spectrometer commissioning, detector performance validation, CMS silicon sensor testing, module QA, metrology, noise testing, detector operations.
- Simulation and Modeling: GEANT4 simulation, Monte Carlo production, detector response studies, event generator workflows, transport codes, physics analysis validation
- Software Tools: C++, Python, ROOT, GEANT4, Git, GitLab, GitHub, JIRA, Linux based analysis workflows, LaTeX, MATLAB, Mathematica.

## Awards and Recognition

- [sPHENIX Hero, June 2026](#). Recognized by the sPHENIX Collaboration for significant contributions to detector tracking calibration and data analysis at BNL.

## Work Experience

**Oct. 2024 – Present**

**Postdoctoral Fellow**

Department of Physics & Astronomy, Wayne State University

- Developed sPHENIX detector analysis workflows for TPC calibration, track based reconstruction studies, strange hadron reconstruction (KFParticle), and detector performance validation.
- Produced GEANT4 simulation samples and validated detector response using reconstruction level observables and data quality monitoring outputs.
- Built ROOT, C++, and Python based analysis pipelines for large scale collider data, simulation production, and physics performance studies.
- Applied machine learning methods to event level observables, model validation, and detector performance studies.

- Served in detector operation and shift leadership roles at BNL, supporting real time detector monitoring and data taking.

**Nov. 2022 – Sep-2024                      Postdoctoral Researcher**

*Center for Cosmology, Particle Physics and Phenomenology (CP3), UCLouvain, Louvain-La-Neuve (Belgium)*

- Led commissioning, installation, calibration, and expert support for the NA62 GTK beam spectrometer at CERN.
- Maintained detector performance validation workflows using data quality monitoring software and technical reporting.
- Supported Run II detector operations, working group activities, and analysis framework development for rare kaon decay studies.
- Contributed to GTK R&D activities and detector documentation for collaboration review and operation support.

**March. 2021 – Oct. 2022                      Senior Scientific Officer**

*National Center for Physics, Islamabad (Pakistan)*

- Contributed to CMS outer tracker R&D through silicon sensor qualification, module assembly support, metrology, QA, and clean room testing.
- Performed module noise testing at different temperatures and supported detector quality assurance after assembly steps.
- Developed a Python based metrology method to estimate module rotation angles and validate detector assembly precision.
- Prepared technical documentation, procurement plans, and proposals development for detector laboratory development.

**July 2019 – Feb. 2021                      Joint-Postdoctoral Researcher**

*Central China Normal University, Wuhan and Stony Brook University, USA*

- Worked as an independent researcher and data analyst for the STAR experiment at BNL, developing reconstruction techniques for strange hadron measurements, and Monte Carlo production coordination.
- Co-supervised one Ph.D. student, prepared funding proposals, presented results at conferences, and published research outcomes in peer-reviewed journals.

## Education And Training

**PhD in Physics (Nuc-Ex), Tsinghua University, Beijing, China                      2014 – 2019**

**MS in Physics, International Islamic University, Islamabad, Pakistan                      2010 – 2012**

**MSc in Physics, CHEP, University of the Punjab, Lahore, Pakistan                      2008 – 2010**

## Research and Innovative Experience

- Full list of publications can be found at: (Inspire-HEP) <https://bit.ly/43gYajm>

## Detector Leadership and Collaboration

- Coordination of the software workgroup and detector calibration group for the fixed-target run of the STAR experiment at RHIC
- Organized international schools and workshops
- Good organizational skills obtained on-call expert team responsible for the GTK beam spectrometer of the NA62 experiment.
- Active contributor to international research collaborations at BNL, and RHIC.